MTH 131(B) Dr. James Ashe
Exam 2 Review Monday 30$^{\text{th}}$ September, 2013

Solve a radical. ex. $\sqrt{x-1}+2=5$**.**

1. Get the radical by itself,

$$\begin{aligned}\sqrt{x-1}+2 &= 5\\ \sqrt{x-1} &= 3\end{aligned}$$

2. Use a square root, simplify, and solve.

$$\begin{aligned}\left(\sqrt{x-1}\right)^2 &= 3^2\\ x-1 &= 9\\ x &= 10\end{aligned}$$

3. Check! $\sqrt{10-1}=9\checkmark$

Solve an equation with rational exponents. ex. $(x-7)^{3/2}-64=0$**.**

1. Get the exponent by itself.

$$(x-7)^{3/2} = 64.$$

2. Get rid of the exponent, using the rule $(b^n)^m=b^{nm}$. Raise both sides by the reciprocal of the exponent ("flip it").

$$\begin{aligned}\left((x-7)^{3/2}\right)^{2/3} &= (64)^{2/3}.\\ x-7 &= 64^{2/3}\\ x &= 16+7\\ x &= 23\end{aligned}$$

3. Check. $(23-7)^{3/2}=64.\checkmark$

Solve the absolute value or indicate that is has no solution. ex. $|x+2|-2=10$

1. Get the absolute value by itself.

$$\begin{aligned}|x+2|-2 &= 10\\ |x+2| &= 12\end{aligned}$$

2. Solve for $+$ and $-$.

$$\begin{aligned}x+2=-12 &\quad\text{and}\quad x+2=12\\ x=-14 &\qquad\qquad\; x=10\end{aligned}$$

3. Check! $|-14+2|-2=10\checkmark$ and $|10+2|-2=10\checkmark$

Solve by factoring. ex. $75x-25=3x^3-x^2$

1. Set the equation equal to 0 and write in descending order.

$$-3x^3+x^2+75x-25=0$$

2. Factor by grouping since there are four unlike terms.

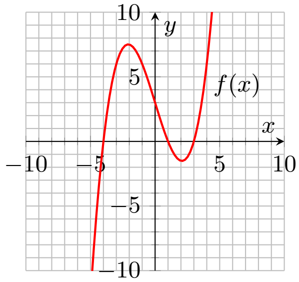
$$-x^2(3x-1)+25(3x-1) = 0$$

3. Factor and solve for each factor.

$$\begin{aligned}(3x-1)(25-x^2) &= 0\\ (3x-1)(5-x)(5+x) &= 0\\ &\Rightarrow x={}^1\!/_3,-5,5\end{aligned}$$

Find the x and y intercepts from a graph.

1. Find all the times the graph intersects the $x-$axis. These are your $x-$intercepts.
2. Find all the times the graph intersects the $y-$axis. These are your $y-$intercepts.
ex.



Find the Domain and the Range.

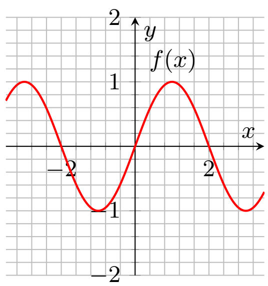
1. Domain= $x's$; Range= $y's$.
2. From a table or list, just find the $x's$ and $y's$. ex.

$$\{(1,2),(0,3),(-1,4),(-2,5)\}$$

$$\begin{aligned} \text{domain}= & \quad \{1,0,-1,-2\} \\ \text{range}= & \quad \{2,3,4,5\} \end{aligned}$$

3. From a graph, look at the horizontal values for the domain $(x's)$ and the vertical values for the range $(y's)$. If they appear to go of the graph, then use $\pm\infty$.

ex.



Determine if the relation is a function. ex. $\{(1,-2),(2,-1),(2,0),(3,1)\}$

1. Check that each x , in (x,y), is paired with only one y value.

$$\begin{aligned} 1 & \rightarrow -2 \\ 2 & \rightarrow -1 \\ 2 & \rightarrow 0 \\ 3 & \rightarrow 1 \end{aligned}$$

2. If any x is paired with two different $y's$ then the relation is NOT a function.

$$\begin{aligned} 2 & \rightarrow -1 \\ 2 & \rightarrow 0 \end{aligned}$$

So this is NOT a function.

3. Be suspicious if an x appears more than once.

Find the slope passing through the given points. ex, $(1,3)$ and $(0,5)$

1. Use the formula, $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Identify (x_1, y_1) and (x_2, y_2).

$$x_1 = 1,\ y_1 = 3 \text{ and } x_2 = 0,\ y_2 = 5$$

2. Plug the values into the formula.

$$m = \frac{5-3}{0-1} = \frac{2}{-1} = -2$$

Use the information to write an equation for the line in point-slope and slope-intercept form. ex. Slope of -2, passing through $(1,3)$.

1. Find the slope: Use the formula, $y - y_1 = m(x - x_1)$. Identify m, x_1,and y_1.

$$\text{The slope is } m = 2\text{, and the point gives } x_1 = 1\text{and } y_1 = 3.$$

2. Plug the values into the formula.

$$y - 3 = -2(x-1) \text{ point-slope form}$$

3. Solve for y to get $y = mx + b$.

$$\begin{aligned} y &= -2x + 2 + 3 \\ y &= -2x + 5 \text{ slope-intercept form} \end{aligned}$$

Use the information to write an equation for the line in point-slope and slope-intercept form. ex. The line perpendicular to $y = -{}^{1}/_{4}x + 2$ and passing through the point $(1, -1)$.

1. Find the slope: $m = -(-{}^{4}/_{1}) = 4$.
It is perpendicular so take the negative reciprocal. (If it was parallel we would just take the slope, $m = -{}^{1}/_{4}$).

2. Plug the values into the formula.
$y - (-1) = 4(x-1) \Rightarrow y + 1 = 4(x-1)$ point-slope form.

3. Solve for y to get $y = mx + b$.

$$\begin{aligned} y &= 4x - 4 - 1 \\ y &= 4x - 5 \text{ slope-intercept form} \end{aligned}$$

Evaluate a function at a given value. ex. $f(x) = 3x^2 - x$ find $f(-2)$ and $f(x+1)$.

1. Plug the given value where there is an x.

$$\begin{aligned} f(-2) &= 3(-2)^2 - (-2) \\ &= 3 \cdot 4 + 2 = 14 \end{aligned}$$

$$\begin{aligned} f(x+1) &= 3(x+1)^2 - (x+1) \\ &= 3(x^2 + 2x + 1) - x - 1 \\ &= 3x^2 + 5x + 2 \end{aligned}$$

Determine if a function is even, odd, or neither. ex $f(x) = 3x^2 - x$, $g(x) = 5x^3 - x$, and $h(x) = 4x^2 - 4$.

Look at the graph to check symmetry or check algebraically.

$f(x)$ is neither since $3x^2$ is even but $-x$ is odd.

$g(x)$ is odd and $h(x)$ is even since each term is odd and even respectively.